

National University of Computer and Emerging Sciences (NUCES), Lahore Campus



Course:	Physics for Engineers	Course Code:	NS1002
Program:	BS(EE)	Semester:	Fall 2022
Duration:	60 minutes	Total Marks:	30
Date:	12-11-2022	Weight:	15%
Section(s):	All	Page(s):	3
Exam:	Midterm 2	Section	
Name:		Roll No.	

Instructions/Notes:

Attempt all questions. Don't forget to write your name, roll number and section.
Write your answer within the space provided only.
You can take a rough sheet, but do not attach it to the paper.
Constant: permittivity of free space: $8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$

Question No.	1	2	Total
Maximum Marks	10(6%)	20(9%)	
Marks Obtained			

Question 1: CLO # 2 (Max. Marks: 10)

1(a): The particle with positive charge q produces an electric field E and an electric potential V at point P . Find the electric potential by moving a test charge q_0 from P to infinity. The test charge is shown at distance r from the particle, during differential displacement ds .

$$\vec{E} \cdot d\vec{s} = E \int ds$$

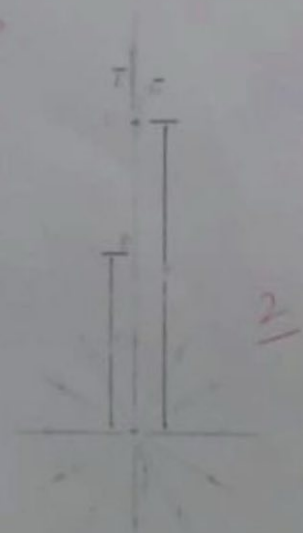
$$V_f - V_i = - \int_R^\infty E \, dr$$

$$V_f = 0, E = \frac{q}{4\pi\epsilon_0 r^2}$$

$$V = - \frac{q}{4\pi\epsilon_0} \int_R^\infty \frac{1}{r^2} \, dr$$

$$V = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_R^\infty$$

$$V = \frac{1}{4\pi\epsilon_0 R}$$



1(b): Over a certain region of space, the electric potential is $V = 5x - 3x^2y + 2yz^2$. Find the expressions for the x , y , and z components of the electric field over this region. What is the magnitude of the electric field at the point 'P' that has coordinates $(1, 0, -2)$ m?

$$E_x = - \frac{\partial V}{\partial x} = -5 + 6xy = -5 + 6(1)(0) = -5 \text{ N/C}$$

$$E_y = - \frac{\partial V}{\partial y} = 3x^2 - 2z^2 = 3(1)^2 - 2(-2)^2 = -5 \text{ N/C}$$

$$E_z = - \frac{\partial V}{\partial z} = -4yz = -4(0)(-2) = 0 \text{ N/C}$$

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$$E = \sqrt{(-5)^2 + (-5)^2 + 0^2} = 7.07 \text{ N/C}$$

Question 2: CLO # 2 (Max. Marks: 20)

2(a): Derive an expression of Capacitance in the case of a spherical capacitor with plate radii 'a' and 'b'. Use Gauss's law for the calculations. Also, draw an illustration.

$$Q = \epsilon_0 E A = \epsilon_0 E (4\pi r^2)$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$V = \int_a^b E dr = -\frac{Q}{4\pi\epsilon_0} \int_a^b \frac{dr}{r^2}$$

$$= -\frac{Q}{4\pi\epsilon_0} \left[\frac{r^{-2+1}}{-2+1} \right]_a^b$$

$$= \frac{Q}{4\pi\epsilon_0} \left[\frac{b-a}{ab} \right]$$

$$C = \frac{Q}{V} = 4\pi\epsilon_0 \frac{ab}{b-a}$$

2(b): A parallel plate capacitor is constructed using a dielectric material whose dielectric constant is 3.00 and whose dielectric strength is 2.00×10^8 V/m. The desired capacitance is $0.250 \mu\text{F}$, and the capacitor must withstand a maximum potential difference of 4000 volts. Find the minimum area of the capacitor plates.

$$E_{\text{max}} = 2 \times 10^8 \text{ V/m} = \frac{\Delta V_{\text{max}}}{d} \quad d = \frac{\Delta V_{\text{max}}}{E_{\text{max}}}$$

$$C = \frac{\kappa \epsilon_0 A}{d} = 0.250 \times 10^{-6} \text{ F} \quad \kappa = 3$$

$$A = \frac{C d}{\kappa \epsilon_0} = \frac{C \Delta V_{\text{max}}}{\kappa \epsilon_0 E_{\text{max}}} = \frac{0.250 \times 10^{-6} \times 4000}{3 \times 8.85 \times 10^{-12} \times 2 \times 10^8}$$

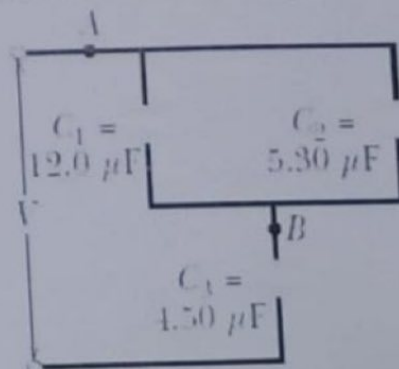
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$$A = 0.188 \text{ m}^2$$

2(c): (i) Find the equivalent capacitance for the combination of Capacitances shown in Figure below, across which potential difference V is applied. Assume

$$C_1 = 12.0 \mu\text{F}, C_2 = 5.30 \mu\text{F}, \text{ and } C_3 = 4.50 \mu\text{F}.$$

(ii) The potential difference applied to the input terminals in Figure is $V = 12.5 \text{ V}$. What is the charge on C_1 ?



$$C_{12} = C_1 + C_2 = 12 + 5.30 = 17.3 \mu\text{F}$$

$$\frac{1}{C_{123}} = \frac{1}{C_{12}} + \frac{1}{C_3}$$

$$= \frac{1}{17.3} + \frac{1}{4.5} = 0.280 \mu\text{F}^{-1}$$

$$C_{23} = 3.57 \mu\text{F}$$

$$q_{123} = C_{123} V = 3.57 \times 12.5 = 44.6 \mu\text{C}$$

$$V_{12} = \frac{q_{12}}{C_{12}} = \frac{44.6 \mu\text{C}}{17.3 \mu\text{F}} = 2.58 \text{ volts}$$

$$q_1 = C_1 V_1 = 12 \mu\text{F} \times 2.58 = 31 \mu\text{C}$$